

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)****Y25 - 2026-2027 Trimester - 4****Project Abstract Submission Form**

Sec No-10	Team No- 4	Project Title- Sign Language Translator App
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1.Abstract :

Communication barriers between hearing-impaired individuals and the wider community remain a persistent challenge, particularly in everyday settings where a human interpreter is not readily available. The proposed Sign Language Translator App aims to bridge this gap by using real-time computer vision and on-device machine learning to translate hand gestures of sign language into readable text and audible speech, enabling more natural and immediate communication.

The application is built natively for Android using Kotlin, with CameraX providing efficient, lifecycle-aware access to the device camera for continuous gesture capture. MediaPipe Hands is used to detect and extract precise hand-landmark coordinates in real time, which are then passed to a TensorFlow Lite gesture-classification model optimized for fast, low-latency, on-device inference. Recognized gestures are converted into text and rendered as natural speech using Android's Text-to-Speech engine, allowing for a fluid, hands-free translation experience without dependence on a constant network connection.

To extend the app beyond a purely offline tool, a lightweight backend service built with Node.js and Express, backed by MongoDB in place of a traditional relational database, is used to power the Phrase History and Model Update modules. MongoDB's flexible, document-oriented structure is well suited to storing variable-length translation logs, per-user phrase history, and versioned model metadata without rigid schema constraints, while also simplifying incremental updates as new gesture classes or model improvements are rolled out to users over time.

Core functional modules include Gesture Capture, responsible for real-time video frame acquisition; Landmark Extraction, which isolates key hand-pose features; a Gesture Classification Model for interpreting signed input; Text & Speech Output for delivering translated results; Phrase History, which persists past translations to MongoDB for later review; and a Model Update Module that fetches improved or expanded classification models from the cloud backend as they become available.

By combining efficient on-device inference with a scalable, document-oriented cloud backend, the system delivers a responsive, privacy-conscious translation experience while still supporting historical logging and continuous model improvement. This project demonstrates the practical integration of mobile computer vision, edge machine learning, and NoSQL-backed cloud services in addressing a meaningful accessibility problem.

2.Team Details

Roll Number(s)	Name of the Student	Signature
2520090018	Rasagnya M	
2520090072	Niyathi K	
2520080016	Varshith Comple	

3.Frameworks / Platforms / Software Used

- Android (Kotlin)
- CameraX
- MediaPipe Hands
- TensorFlow Lite
- Text-to-Speech
- MongoDB (cloud sync backend)
- Node.js/Express REST API

Guide Name and Signature (remarks if any) _____